RaagSense

AI-Based Hindustani Raag Recognition

Group Member: Aarush Ram Anandh | Individual Project

Project Description

RaagSense will take a short audio recording of Hindustani classical music and estimate the probability that it belongs to each raag known by the system. Identifying a raag requires understanding characteristic melodic movements rather than simply checking which notes are present. The project will explore whether an AI model can learn these patterns from labelled recordings.

Datasets

The project will use the Saraga Hindustani dataset and the CompMusic Hindustani Music Dataset. These datasets contain Hindustani classical recordings, raag labels, tonic information, pitch features, and other musical annotations.

AI Concepts and Techniques

- Audio preprocessing and segmentation
- Pitch and tonic detection
- Tonic-normalized melodic representations
- Spectrogram or constant-Q audio features
- CNN-LSTM or a small transformer model
- Multiclass probability prediction

Implementation Plan

The recordings will be divided into short excerpts and converted into representations of their pitch and melodic movement. These representations will be normalized relative to the tonic and used to train a classification model. The model will learn patterns associated with different raags and produce a probability score for each supported raag.

Expected Input and Output

The input will be a short audio recording of Hindustani classical music. The output will be a ranked list of possible raags with probability scores.

Expected Final Features

- Upload and process an audio file
- Detect the tonic and melodic pitch contour
- Predict the most likely raags and display their probability scores
- Visualize the extracted pitch movement
- Test the system using unseen recordings

AI Component

The AI component is the classification model that learns characteristic melodic patterns from labelled recordings and uses them to classify previously unseen audio excerpts.